Original Article

Novel three-dimensional methods to analyze the morphology of the nasal cavity and pharyngeal airway

Xiaowen Niu[a]; Sivaranjani Madhan[a]; Marie A. Cornelis[b]; Paolo M. Cattaneo[b]

ABSTRACT

Objectives: To assess the intraexaminer and interexaminer reliabilities of novel semiautomatic methods to segment the nasal cavity (NC) and pharyngeal airway (PA) and to determine the minimal cross-sectional area (CS) and hydraulic diameter (HD) of the PA.

Materials and Methods: To test reproducibility, two examiners analyzed the NC and PA independently in 10 retrospectively selected cone beam computed tomography (CBCT) images using semiautomatic segmentation. The PA centerline was determined to assess the minimal CS and HD. The intraclass correlation coefficient (ICC) was used to calculate intraexaminer and interexaminer reliabilities. Measurement errors were assessed by Dahlberg's formula and paired *t*-tests. The level of agreement was assessed using the Bland-Altman method.

Results: Intraexaminer and interexaminer reliabilities were excellent (minimal ICC, 0.960). The error of the method was good except for interexaminer values for the oropharynx ($P = .016$). The minimal CS and HD measurements were reliable (minimal ICC, 0.993; narrow limits of agreement).

Conclusions: The novel methods for analysis of the NC and PA are reliable. The minimal CS and HD demonstrated excellent reliabilities, which are critical to detect the most constricted part of the PA. Separation of the oropharynx from the voids close to the retroglossal area is not trivial and should be considered with caution. (*Angle Orthod.* 2021;91:320–328.)

KEY WORDS: Orthodontics; Three-dimensional; Nasal cavity; Pharyngeal airways

INTRODUCTION

The nasal cavity (NC) and pharyngeal airway (PA) constitute the beginning of the upper respiratory tract (URT). It is understood that the definite diagnosis of sleep-disordered breathing (SDB) should be done only by a sleep physician. On the other hand, for diagnostic purposes, orthodontists regularly acquire radiographic records of the craniofacial structures that usually encompass both the NC and the PA. In addition, orthodontists have a broad patient population, with contact maintained over a long time; they are therefore in a unique position to screen for SDB, as clearly emphasized recently.[1]

Cone beam computed tomography (CBCT) is becoming the first choice for three-dimensional (3D) imaging in dentistry because it offers lower costs and less radiation exposure than medical computed tomography (CT).[2] CBCT provides good contrast between soft tissues and air,[3] potentially offering an accurate representation of the NC and PA. Thus, finding fast and robust tools for investigating the morphology of the NC and PA based on orthodontic 3D radiological records has significant potential. However, several factors may compromise the quality of the assessment.[2–4]

The NC is surrounded by nasal turbinates and four paired, air-filled paranasal sinuses, which renders its segmentation particularly difficult. Segmentation of the NC can be done either manually, semiautomatically, or automatically. Manual segmentation has been described as accurate but very time consuming (up to 10 hours for a data set), whereas the semiautomatic

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method was shown to be faster.[5] Automatic segmentation performed using a general threshold is fast; however, it may not be optimal because some parts of the NC are very narrow, and the complex boundary of the NC makes it difficult to discriminate the sinuses from the NC.[3] Studies suggested using reference planes to separate the NC from the paranasal sinuses, yet the results associated with the NC were not accurate because the complex anatomic boundaries cannot be depicted using simple planes.[6]

Segmentation of the PA in CBCT is not complicated per se and has been described in several articles.[3,7–9] A systematic review concluded that volumetric measurement of the PA and NC on CBCT scans is the most common approach, demonstrating higher reliability than estimation of the cross-sectional area (CS).[8] Also, the procedure to select the threshold value(s) for the segmentation procedure is a critical process, especially on CBCTs, as the degree of airway filling is directly related to the chosen threshold, yet most of the studies provided no details of the procedure applied. Therefore, the reported moderate to excellent reliability for volumetric measurements should be considered with caution.

For patients with suspected URT obstruction, measuring the width, anteroposterior length, or partial and total volumes may not be sufficient to depict its complex morphology. Identification of the restricted section of the air passage, the related area, and shape are more relevant so that parameters such as minimal CS and minimal hydraulic diameter (HD) would be of particular importance.

Current approaches to segment the NC are either time consuming or not accurate in separating the NC from the surrounding tissues and sinuses. In addition, no study has been conducted assessing the minimal CS and HD of the PA perpendicularly to the PA unfolding. Therefore, this technical report aimed to assess the validity and the intraexaminer and interexaminer reliabilities of all the steps involved in the novel semiautomatic methods to segment both the NC and PA volumes and to determine the minimal CS and the minimal HD of the PA. The hypothesis was that the proposed methods to segment the NC and PA were reliable.

MATERIALS AND METHODS

The sample size was determined using the method developed for reliability studies[10] using interexaminer reliability (The null hypothesis H0, $\rho_0 = 0.5$ [minimally acceptable level of reliability], and the alternative hypothesis H1, $\rho_1 = 0.9$ [expected level of reliability]; $\alpha = 0.05$; $\beta = 0.20$; number of replicates =2). Based on these parameters, nine observations were needed. To avoid selection bias, ten fully anonymized, large field-of-view CBCTs comprising the cranial base, maxilla, mandible, and first four cervical vertebrae and associated airway were randomly selected (random.org) without limitation as to sex and age from the clinical database of the Section of Orthodontics, Department of Dentistry and Oral Health, Aarhus University, Denmark (all CBCTs were taken for specific diagnostic purposes). The exclusion criteria were significant nasal issues (ie, enlarged turbinates and nasal constriction), swallowing during scan acquisition, craniofacial deformity, and posture and movement artefacts. The study was reported and the data were analyzed with the permission of the Danish Data Protection Agency (Reference 2016-051-000001, Serial No. 1393).

All CBCTs were taken with a supine scanner (NewTom 5G, QR, Verona, Italy; 110 kVp, 5 mA, 0.3 mm isotropic voxel, 18 seconds scanning time, 3.6 second exposure time, 18 × 16 cm field of view). The scanning protocol provided that the subjects had their heads resting without any head positioner in centric occlusion with their lips and tongues in a resting position. CBCT data were imported through the DICOM format into Mimics 21 software (Materialise, Leuven, Belgium).

One orthodontist (Dr Niu) and one oral maxillofacial radiologist (Dr Madhan) were calibrated for measuring the NC and PA on other CBCTs than the ones used in the present study under supervision (Dr Cattaneo, with extensive expertise in 3D analysis). After calibration, the analyses were performed independently by the two examiners for all patients. Subsequently, one examiner (Dr Niu) repeated the analyses after 2 weeks while blind to the previous assessments.

The UTR region was defined as the space extending from the plane passing through Ntip and ANS to the epiglottis (E). The UTR was further divided into the NC and PA using the PNS-So plane (for the definitions of abbreviations, landmarks, and planes, see Table 1; Figure 1). The maxillary, frontal, ethmoid, and sphenoid sinuses were not included.

As the gray value scale among the different CBCTs is not constant,[4] for each CBCT, two different thresholds were determined using specific profile lines: for the NC, on the coronal view, a profile line was traced from the left to right maxillary sinuses (Figure 2A); for the PA, a profile line was traced on the midsagittal image from Ba to PNS (Figure 2C). The appropriate threshold values were determined using the "four-point" method applied to the profile lines histogram, as previously described (Figure 2B,D).[7,9]

To segment the NC, a layer (called a "mask") with the relevant structures was defined and color coded using the appropriate threshold value. The mask was first cleared and then the airway space was roughly

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Table 1. Landmarks Selected for Airway Analysis

Parameter	Description
Point	
ANS	The most anterior point on the nasal spine
Ba	The most posteroinferior point on the clivus
E	Most superior point of the epiglottis
MoL	The distal-palatal tip of the first left molar in the upper jaw
MoR	The distal-palatal tip of the first right molar in the upper jaw
N	The intersection of the internasal and frontonasal sutures in the midsagittal plane
Ntip	The tip of the nasal bone
OrL	Orbital left, the most inferior anterior point on left orbit's margin
OrR	Orbital right, the most inferior anterior point on right orbit's margin
PNS	The most posterior point on the nasal spine
PoL	Porion left: the most upper point on the left bony external auditory meatus
PoR	Porion left: the most upper point on right bony external auditory meatus
S	The midpoint of the sella turcica
So	The midpoint of the sella-basion line
ii	The point midway between the incisal edges of the maxillary central incisors
Reference plane	
Frankfurt plane	Plane passing through the inferior borders of the bony orbits, encompassed by OrR and OrL and the upper margin of the auditory meatus encompassed by PoR and PoL
Sagittal SN plane	Plane perpendicular to Frankfurt plane passing through the S and N points
NTip-ANS plane	Plane through NTip and ANS points, perpendicular to the sagittal SN plane
PNS-So plane	Plane through PNS and So points, perpendicular to the sagittal SN plane
PNS-Ba plane	Plane through PNS and Ba points, perpendicular to the sagittal SN plane
Occlusion plane	Plane through MoL, MoR, and ii points
E plane	Plane through E point, parallel to the Frankfurt plane

painted every 10th axial image; the same procedure was repeated for the coronal views. This editing produced a checkerboard-type pattern mask (Figure 3). The "smart expand" tool was then applied to create the NC mask followed by minor adjustments if needed. The 3D volumetric object depicting the NC was then calculated (Figure 3).

To segment the PA, a mask was created using the appropriate threshold, and the corresponding 3D

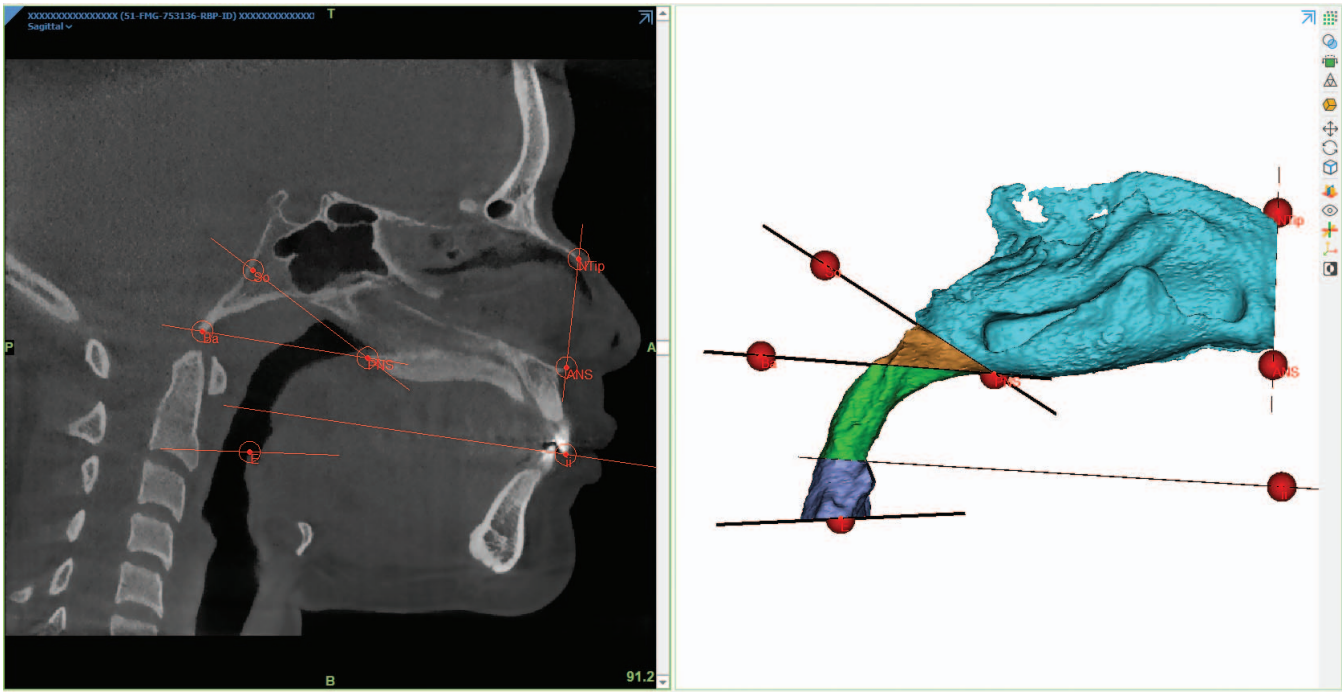


Figure 1. Three-dimensional skull reconstruction: nasal cavity (blue), lower nasopharynx (brown), velopharynx (green), and oropharynx (purple).

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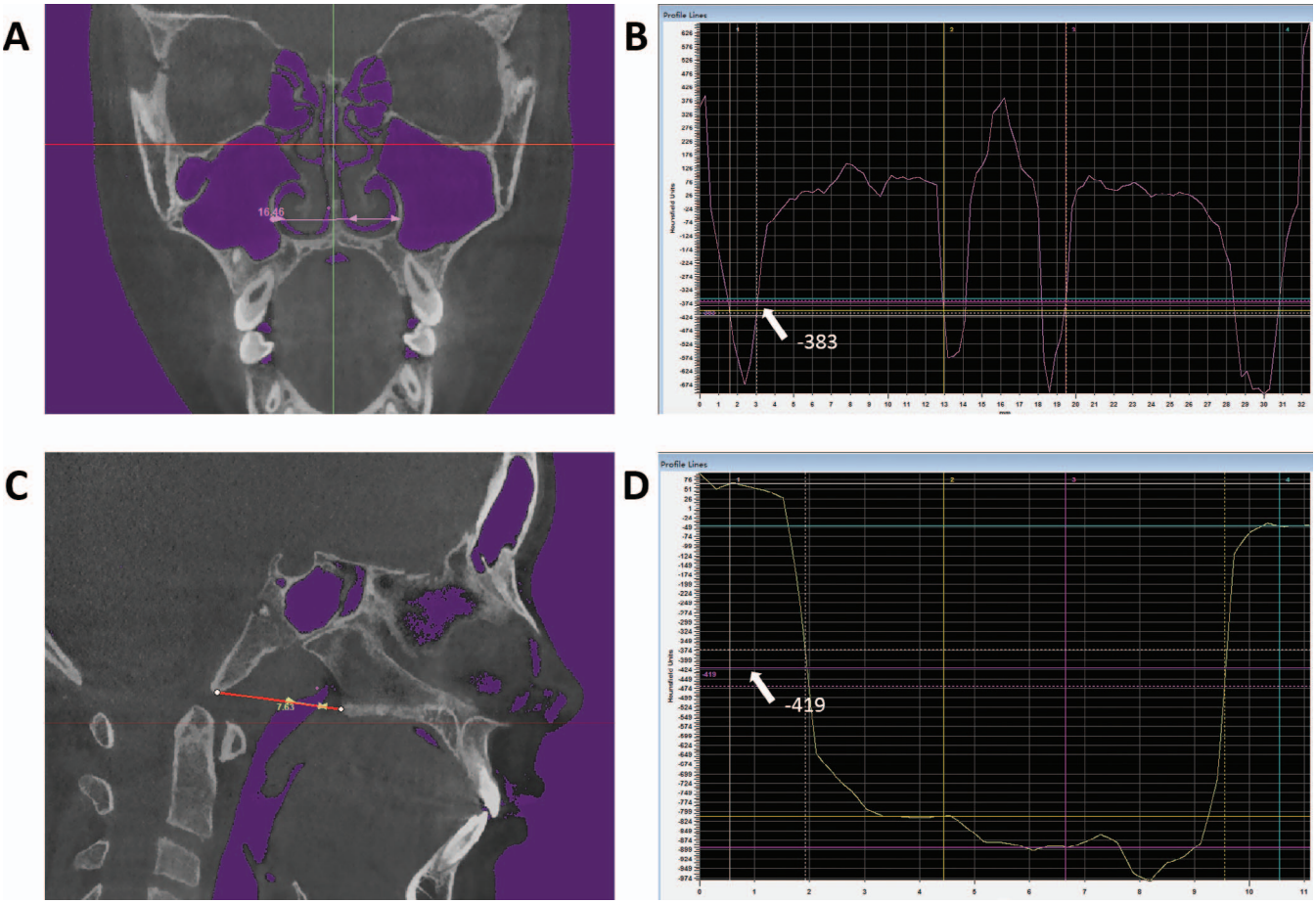


Figure 2. (A) Profile line for nasal cavity. (B) Histogram graph for NC. (C) Profile line for airway. (D) Histogram graph for the PA.

model was generated. The 3D model of the craniofacial structures was then calculated.[7,9]

During the process to extract and determine the centerline of the PA, to avoid the generation of multiple mainlines and a number of incorrect side branches, given that the 3D model of the PA is far from being a smooth pipe, the centerline default settings used by the algorithm had to be modified manually. This decreased the level of details considered in the extraction process, thus obtaining a centerline with only the mainline (Figure 4), with control points set about 1 mm apart.

Two reference planes were defined (Table 1): the Frankfort plane, used as the horizontal plane, and the

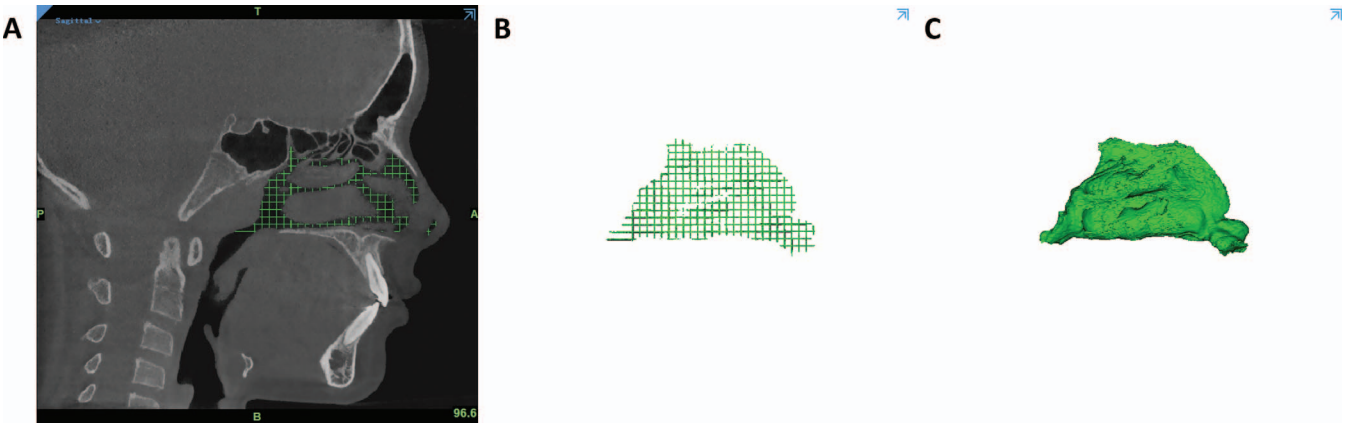


Figure 3. (A) Initial checkerboard-type pattern mask. (B) Three-dimensional reconstruction of the checkerboard type. (C) Three-dimensional reconstruction of the NC after smart expansion.

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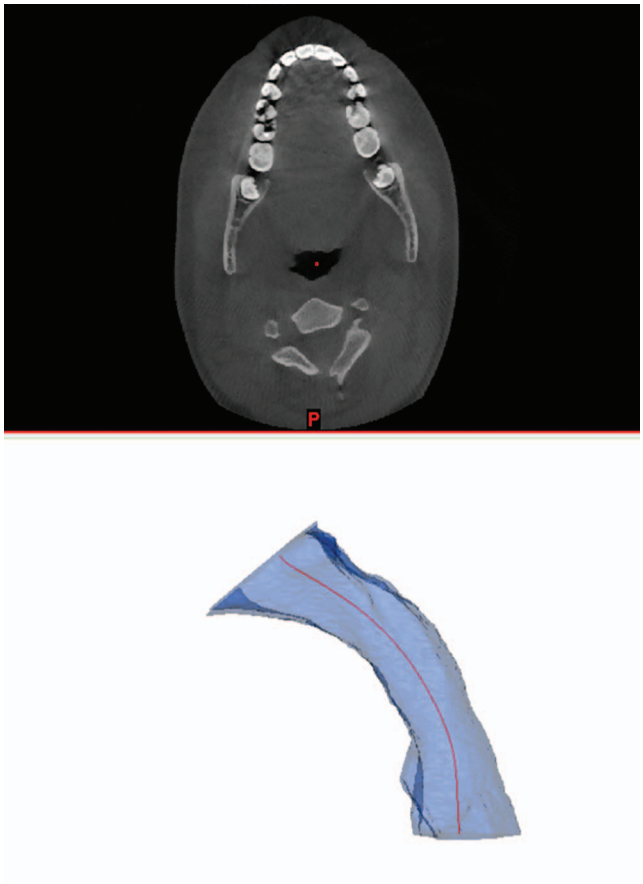


Figure 4. Centerline for the pharyngeal airway.

sagittal SN plane, as the sagittal plane. The total PA was divided into three parts according to the location of four planes (PNS-So, PNS-Ba, occlusion, E plane), and their volumes were assessed (Table 2, Figure 1). Using the centerline of the PA, the CS and HD could be assessed perpendicularly to the centerline.

Statistical Analysis

Intraexaminer and interexaminer reliabilities were calculated using the interclass correlation coefficient (ICC), with 95% confidence intervals for the measurements obtained by the two examiners and for both assessment periods (SPSS 26.0, IBM, Armonk, N.Y.). Normality was checked with the Kolmogorov Smirnov test. Systematic intraexaminer and interexaminer errors were tested. Because the data were normally distributed, paired *t*-tests were used. The technical error of measurements at the interobserver and intraobserver levels was tested with the Dahlberg formula, whereas measurement agreement was assessed with the Bland-Altman method (means and limits of agreement: 95% limit of agreement [LoA] = 1.96 × standard deviation [SD]).[11]

Table 2. NC and PA Volumes and Cross-Sections

Parameter	Description
Volume measurement	
NC volume	Bounded anteriorly by the NTip-ANS plane and posteriorly by the PNS-So plane
Nasopharynx volume	Bounded superiorly by the PNS-So plane and inferiorly by the PNS-Ba plane
Velopharynx volume	Bounded superiorly by the PNS-Ba plane and inferiorly by the occlusion plane
Oropharynx volume	Bounded superiorly by occlusion plane and inferiorly by the E1-E2 plane
Total PA volume	Bounded superiorly by PNS-So plane and inferiorly by the E1-E2 plane
Cross-section measurement	
MCS	The minimal cross-section area in total PA
HD	The minimal hydraulic diameter in total PA

RESULTS

The sample mean age was 12.2 ± 2.03 years (range, 9.6–16.6). Differences within intraexaminer and interexaminer measurements were both normally distributed. Descriptive statistics of intraexaminer and interexaminer measurements and *P* values (paired *t*-tests) are reported in Table 3 and Figure S1. A statistically significant interexaminer difference was seen for the oropharynx ($P = .016$), with a measurement bias observed in the Bland-Altman plot (Figure S1).

The Dahlberg values, the percentage of measurement error (with respect to the mean of the first measurement/examiner), and LoA are provided in Table 4. The percentage of error was generally below 6%, whereas the oropharynx displayed an interexaminer error of 12%. The LoAs were generally narrow except for the NC.

The ICC values for the intraexaminer and interexaminer comparisons were above 0.96, indicating excellent reliability (Table 4). The position of the minimal CS and HD was located in the oropharynx in 7 and 6 cases of 10, respectively. The mean values of the minimal CS and HD are reported in Figure 5. The location of the minimal CS and HD did not coincide in 7 of 10 cases, but were both located within the oropharynx.

DISCUSSION

SDB, and obstructive sleep apnea (OSA) in particular, may, if left untreated, decrease the quality of life.[1,12] OSA is suggested to be related to a PA anatomic disorder[13] so that the treatment options are mostly associated with enlarging the PA.[12]

CBCT scans taken for orthodontic purposes represent a low-dose 3D radiological record facilitating the

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Table 3. Means, SDs, and 95% Confidence Intervals (CI) for Intraexaminer and Interexaminer Differences and Paired *t*-Test *P* Values to Detect Systematic Errors

	Intraexaminer (First Examiner)						Interexaminer (First and Second Examiners)					
	First Measurement	Second Measurement	Difference	95% CI		*P*	First Examiner	Second Examiner	Difference	95% CI		*P*
Variable	Mean (SD)	Mean (SD)	Mean (SD)	Upper	Lower	Value	Mean (SD)	Mean (SD)	Mean (SD)	Upper	Lower	Value
NC volume, mm^3	11,205 (2234)	10,877 (2160)	328 (840)	−273	928	0.124	11,205 (2234)	10,904 (2350)	300 (812)	−284	883	.138
Total PA volume, mm^3	8427 (3199)	8466 (3324)	−40 (265)	−229	150	0.648	8427 (3199)	8787 (2956)	−361 (533)	−742	−20	.061
Nasopharynx volume, mm^3	1931 (1007)	1898 (1004)	32 (97)	−37	102	0.314	1931 (1007)	1926 (957)	5 (127)	−85	96	.899
Velopharynx volume, mm^3	2904 (1023)	2948 (1119)	−44 (193)	−182	93	0.485	2904 (1023)	3062 (1154)	−158 (346)	−405	90	.183
Oropharynx volume, mm^3	3586 (1984)	3642 (1982)	−56 (142)	−158	46	0.244	3586 (1984)	4015 (1978)	−429 (457)	−756	−102	.016*
Minimal cross-sectional area, mm^2	86.95 (58.27)	86.52 (57.09)	0.44 (2.49)	−1.34	2.22	0.590	86.95 (58.27)	87.70 (56.85)	0.75 (3.07)	−2.95	1.45	.461
Hydraulic diameter, mm	7.55 (2.38)	7.45 (2.36)	0.10 (0.39)	−0.18	0.39	0.421	7.55 (2.38)	7.80 (2.16)	−0.25 (0.67)	−0.73	0.23	.270

* $P \leq .05$.

assessment of both the PA and NC.[2,3] Therefore, reliable 3D methods to analyze NC and PA morphology are highly desirable.

To date, several authors have reported methods to measure the NC.[6,14,15] However, automatic segmentation methods using commercial dental software were reported to be inaccurate.[16] In this study, the masks were sketched manually every 10 slides and then a semiautomatic procedure was followed. As the initial mask was created manually, the present approach could overcome the limitations of using a general threshold. In addition, the segmentation could be completed within 1 hour, which is considerably less compared to a fully manual segmentation,[5] although comparable with what was reported in the literature for semiautomatic segmentation.[14,15] Volume measurements for the NC demonstrated good correlation coefficients and low measurement errors, indicating excellent intraexaminer and interexaminer reliabilities.

A systematic review of CBCT-based PA measurements reported that the descriptions of the thresholds used were incomplete.[17] Yet, because of the inconsistency of the gray value scale in CBCTs, it is important to determine the threshold for each scan.[4] The PA segmentation approach of this study has been used successfully before,[7,9,18] with the results confirming excellent intraexaminer and interexaminer reliabilities. The only exception was the variation in the interexaminer values of the oropharynx volume. Anatomically, the oropharynx communicates anteriorly with the oral cavity by the facial isthmus, which is bounded superiorly by the soft palate and inferiorly by the tongue. The inconsistency of the current measurements was found to be related to the tongue posture, which can create possible voids close to the retroglossal area. The results of the present study differed from two previously published studies, which reported high reliability for the oropharynx volume.[19,20] The reason may be related to the procedure used to delimit the anterior boundary of the oropharynx: in one study, it was defined by a plane perpendicular to the FH passing through the PNS;[20] in the second one, the plane "extending from PNS to the tip of the epiglottis" was used.[19] Although these approaches were reported as reproducible, their accuracy in the assessment of the true volume of the oropharynx may be heavily influenced by head and tongue posture.

Another systematic review comparing PA size between patients with and without OSA concluded that the minimal CS of the PA was a crucial parameter in the analysis of airway collapsibility.[21] Zimmerman et al. tested the reliability of the measurement method for minimal CS using Dolphin 3D (Dolphin Imaging & Management Solutions, Chatsworth, CA, U.S.A.) and reported that the intraexaminer reliability was moderate

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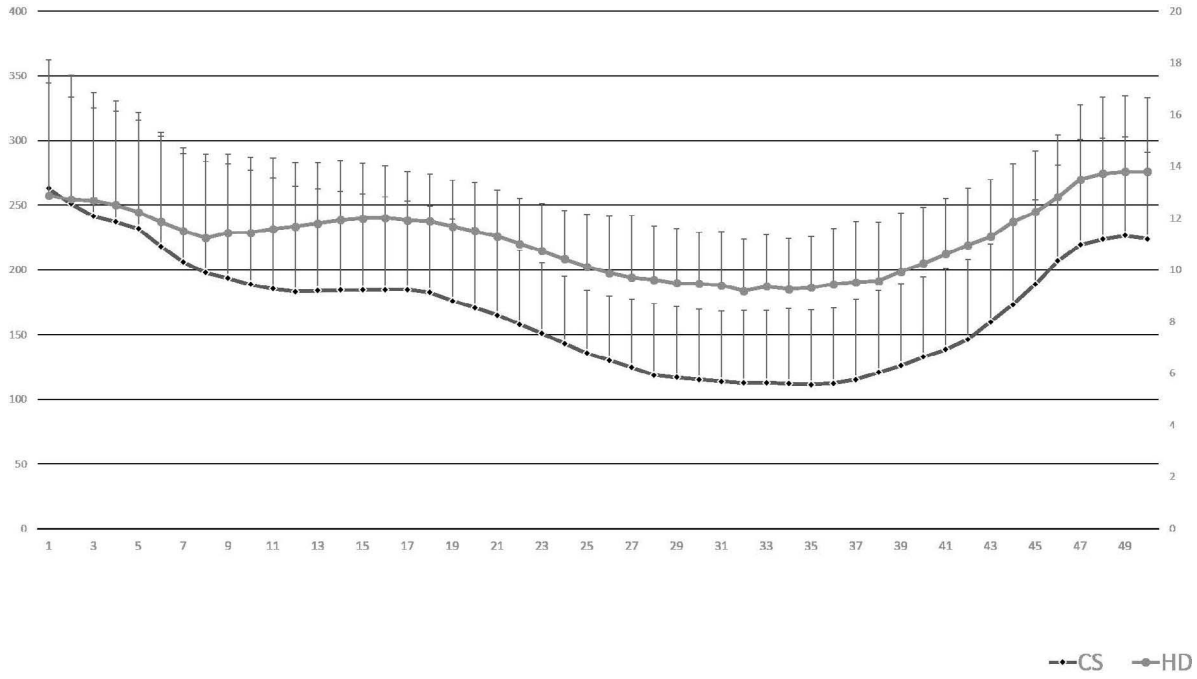


Figure 5. PA morphology analysis. Location of the minimal CS and HD of the PA with respect to control points: mean ± SD.

Table 4. Intraexaminer and Interexaminer Reliabilities Estimated by the Dahlberg Formula, Bland-Altman Plot, and Intervention ICC for Each Measurement

	Intraexaminer				Interexaminer			
Variable	Dahlberg	Percentage[a]	LoA	ICC	Dahlberg	Percentage[a]	LoA	ICC
NC volume, mm³	609	5.4	1646	0.960	587	5.2	1592	0.966
Total PA volume, mm³	180	2.1	508	0.990	439	5.2	1045	0.998
Nasopharynx volume, mm³	69	3.6	190	0.998	85	4.4	249	0.996
Velopharynx volume, mm³	133	4.6	378	0.992	258	8.9	678	0.971
Oropharynx volume, mm³	103	2.9	278	0.999	431	12.0	896	0.976
Minimal cross-sectional area, mm³	1.70	2	4.88	0.998	2.13	2.4	6.02	0.997
Hydraulic diameter, mm	0.27	3.6	0.76	0.997	0.48	6.4	1.31	0.993

[a] Calculated with respect to the first measurement or first examiner.

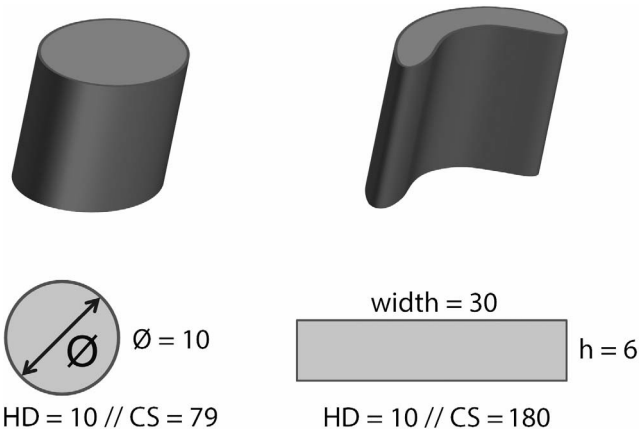


Figure 6. The cross-sectional area of the rectangular shape is more than 2× the circle. However, the hydraulic diameters are identical.

(ICC = 0.704).[19] In most of the studies looking at the most constricted areas,[7,9,19,22] the assessment was performed using cross sections parallel to a user-defined horizontal plane. This could provide misleading results because the PA is characterized by a complex morphology that cannot be assumed to be perpendicular to a given horizontal plane.

In most of the pharyngeal tract, the shape of the lumen does not resemble a circle, but it assumes very jagged forms. As in a pipe the resistance to flow is greatly influenced by its shape, in fluid dynamics, the HD is often used to predict the actual flow in noncircular ducts. In Figure 6, the round and rectangular shapes have the same HD (ie, they allow for the same flow given the same boundary conditions); however, the cross section of the rectangular shape is more than twice the section of the round shape. This

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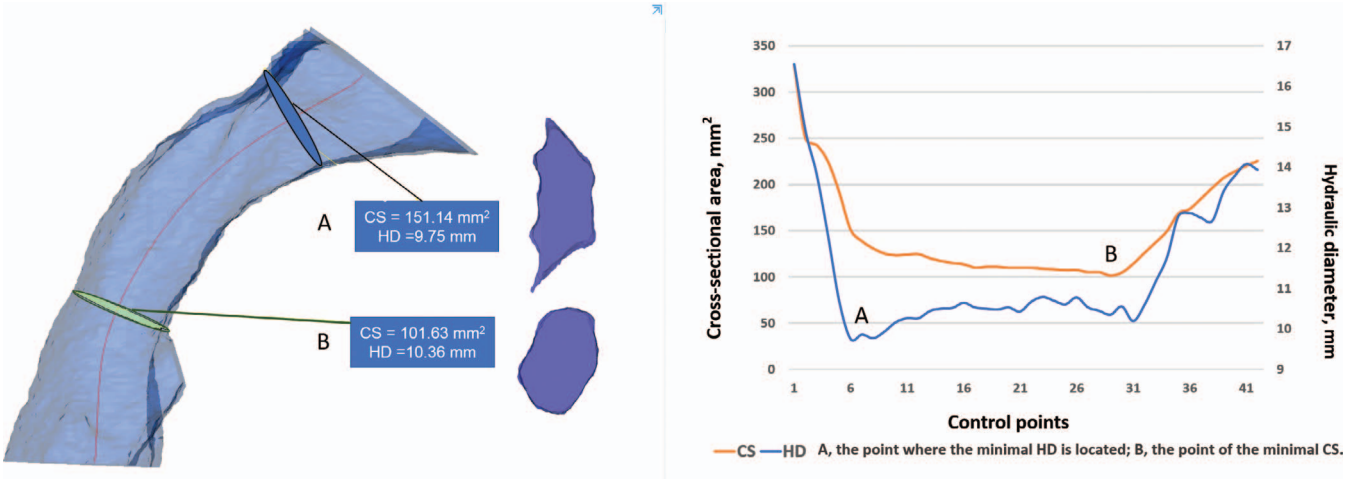


Figure 7. Left: 3D display of a PA and its centerline for one patient. Right: plots of the CS and HD for each control point along the PA.

fact suggests that, for characterizing PA morphology, the HD might be a crucial measure, more so than the CS. In the present study, the PA centerline was determined for each CBCT data set individually following the actual shape of the PA so that at each control point the CS and HD could be assessed perpendicularly to the centerline (and thus to the airflow). Unsurprisingly, given the irregular shape of the PA, the location of the minimal CS and HD were not always coinciding. This is well depicted in Figure 7, where the minimal CS is located in the oropharynx (Figure 7B), whereas the minimal HD is in the nasopharynx (Figure 7A).

This study was the first to propose a method to assess both the minimal CS and HD and their locations along the PA, with both measurements proving to be highly reliable and fast and easy to conduct, thus possible to implement and use in daily clinical practice. This technical article presents novel semiautomatic approaches to analyze the NC and PA morphology and to determine the CS and HD of the PA. The methods were applied to a heterogeneous group of patients scanned in a supine position. These approaches were reliable, with some concerns with respect to the separation of the oropharynx from the retroglossal area.

Although excellent reliability was obtained for most measurements, this was assessed on a limited number of cases. Thus, the accuracy and reliability of this method should be verified in further studies including a larger data set with direct clinical assessment of the URT, for example, using endoscopy. Indeed, quantitative airway evaluation of the URT based on CBCT may not be directly correlated to the presence or development of SDB, as using CBCT only one, static picture is captured, in awake patients.[23]

CONCLUSIONS

- These novel 3D methods to segment and analyze the NC and PA were highly reliable.
- The minimal CS and minimal HD associated with the PA centerline demonstrated excellent reliability, thus they can be used as critical measurements to detect the level of PA constriction.
- The separation of the oropharynx from the voids close to the retroglossal area is not trivial and should be considered with caution.

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SUPPLEMENTAL DATA

Supplemental Figure 1 is available online: Supplemental Figure 1. Bland-Altman plots

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